

Voice or Mask

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Voice or Mask? Stylometric Forensic Analysis of Two Contemporary Nigerian Poets

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Abstract

The era of fluent large language models has destabilised a long-standing assumption in authorship attribution: that a writer's surface features are difficult to imitate at scale. This paper tests that assumption against two contemporary Nigerian poets, Sule Egya (E.E. Sule) and Toyin Shittu, whose published work spans poetry, critical prose, and academic articles from 2009 to 2025. The corpus is public. Sixteen human passages (7,151 words) and sixteen LLM imitations (2,623 words) were produced across two free-tier models (nex-agi/nex-n2.5-mini and nex-nagi/nex-n2.5-pro via OpenRouter). Stylometric features per passage include sentence length distribution, type-token ratio, mean word length, punctuation density, function-word ratio, and a vowel-group syllable estimate. Across both authors the LLM imitations show lower mean syllables per word than the human originals: for Egya 1.65 falls to 1.44, for Shittu 1.94 falls to 1.73. Imitations also show

higher raw type-token ratio (Egya 0.59 to 0.77; Shittu 0.54 to 0.64), but the passages are not length-matched: human passages run roughly 450 words on average against roughly 164 for the imitations, and TTR falls with length, so the TTR gap is confounded and I do not treat it as a finding. The syllable gap survives that concern and holds across both authors and both models. This is a small, length-documented observational note on two authors, not a forensic claim: 16 passages per condition, two LLM providers, temperature 0.01, and a mixed corpus of lyrics, interviews, and prize coverage. A larger, length-matched, multi-model evaluation is future work.

1. Introduction

GPT-3 arrived in 2020. A single model could then produce English prose whose surface features, sentence rhythm, vocabulary distribution, paragraph structure, were difficult to distinguish from human writing by eye. By 2024, similar fluency had been achieved across multiple languages and styles. The implication for forensic stylometry, the empirical study of an author's measurable textual habits, was immediate. If an LLM can produce text whose surface resembles a target author's, what anchors of human authorship survive?

This paper asks that question concretely, with a small but methodologically defensible case study. Two contemporary Nigerian poets. Their published work spans a decade and a half. Against their styles two commercial-grade LLMs are asked to imitate. I treat the question empirically rather than philosophically. I am not asking whether authorship is “real” in some metaphysical sense; I am asking whether the measurable features of an author's text can be distinguished from a plausible machine imitation under realistic conditions.

Four research questions structure the empirical work, summarised below and elaborated in Section 1.1. **RQ1:** Do LLM imitations preserve the lexical-diversity profile of a target author, as measured by type-token ratio? **RQ2:** Do LLM imitations preserve the syllable-length distribution of a target author, as measured by mean syllables per word? **RQ3:** Does the lexical signature, if any, generalise across authors with distinct cultural profiles (literary-cultural vocabulary vs academic-theoretical vocabulary)? **RQ4:** Does the cultural-idiom layer (Igbo and Yoruba terms in Egya; academic vocabulary in Shittu) survive the LLM's surface imitation? Across both authors and both models tested, the answer to RQ2 is no: LLM imitations use shorter, less polysyllabic words than the humans they imitate, and this gap survives the length caveat. The answer to RQ1 is unresolved in this corpus: raw TTR is higher for the imitations, but the conditions differ roughly threefold in passage length and TTR falls with length, so the raw gap is

confounded and no length-matched test was run. The answer to RQ3 is partial: the syllable-length signal generalises across both authors; the lexical-diversity signal cannot be assessed until lengths are matched. The answer to RQ4 is partial: LLMs approximate the surface of culturally embedded vocabulary without fully replicating it.

1.1 Research questions in detail

RQ1: Lexical diversity. Type-token ratio (TTR) is a long-established measure of vocabulary diversity per fixed-length text (Mosteller and Wallace, 1964; Tweedie and Baayen, 1998), with the emphasis on fixed-length: TTR falls mechanically as text grows, so comparing TTR across conditions of different passage length is confounded by construction. LLMs trained on broad web corpora are not, in principle, constrained to the lexical profile of any one author. I report whether, in practice, a prompted LLM’s output sits above or below the target author’s raw TTR, with the length caveat carried explicitly: at this corpus size the conditions are not length-matched, so this question can be posed but not cleanly answered here.

RQ2: Syllable-length distribution. Mean syllables per word captures polysyllabic vocabulary, a stylistic register marker (Biber, 1988). Literary authors and academic authors differ in how polysyllabic their vocabulary tends to be; I test whether LLMs preserve the target author’s profile or substitute shorter, more common words.

RQ3: Cross-author generalisation. A lexical signature that holds for one author might fail on another. I test whether the same lexical features discriminate human from LLM for both authors, despite their distinct cultural and academic profiles.

RQ4: Cultural idiom. Each author’s text carries vocabulary specific to its cultural context. Egba deploys Igbo and Yoruba terms; Shittu uses critical-theoretical vocabulary. I test whether LLMs prompted with a target author’s style replicate these culturally specific vocabularies or revert to a default English-vocabulary profile.

The choice of Nigerian poets is deliberate, not parochial. Nigerian literature in English has a documented tradition of literary criticism that touches on stylistic analysis (Asein 1978). The two authors selected, Sule Egba and Toyin Shittu, both write poetry and non-poetry prose. Both have published pre- and post-COVID-19. Both have public-domain or fair-use excerpts available online. The cultural density of their work is asymmetric and that asymmetry matters for the imitation task. Egba deploys Igbo vocabulary and concepts at high frequency (the terms *chi*, *ala*, *ndi*, *ile*, *ife* appear 47 times across 10 files), with Yoruba terms (*omo*, *ola*, *ifa*) appearing in his critical essays on Nigerian literary history. Shittu draws on Western critical-theoretical vocabulary (representation, narrative, articulation, interrogation) and on Nigerian institutional references (ANA, Association of Nigerian Authors, 2025 shortlist), with no Igbo or Yoruba terms in the

corpus. If LLMs can imitate Egya’s culturally embedded idiom at the surface, they can imitate most English-language literary authors. Shittu’s academic register is a weaker test of cultural imitation, but a stronger test of register imitation.

2. Related Work

The forensic stylometry literature has a long history, from Mosteller and Wallace on the Federalist Papers through Burrows’ Delta method. The post-LLM literature is shorter and largely framed around authorship obfuscation rather than attribution, with notable work on stylometric obfuscation methods (Xing et al., 2024) and on the failure modes and bias of commercial AI-text detectors (Liang et al., 2023; Weber-Wulff et al., 2023). Liang et al. (2023) show that GPT detectors systematically misclassify non-native English writing as AI-generated, with a 61% false-positive rate on TOEFL essays. Weber-Wulff et al. (2023) tested 14 publicly available detection tools and two commercial systems (Turnitin, PlagiarismCheck) and found none of them reliable enough for academic misconduct decisions. I take the position that the more rigorous empirical question, does the imitator’s output leave a measurable signature, is not yet well answered for literary authorship, particularly for non-Anglophone literary traditions.

3. Corpus

3.1 Composition

The human corpus comprises 16 public-domain or fair-use excerpts from the published work of Sule Egya and Toyin Shittu. Excerpts are stored as `.txt` files in `corpus/`, with metadata in `corpus/metadata.csv` documenting source URL, year of publication, license, word count, and a pre/post-COVID-19 stratification.

Author	n passages	n words	Years	Genres
Sule Egya (E.E. Sule)	10	4,137	2009–2022	Poetry, critical essays, interview
Toyin Shittu	6	3,014	2024–2025	Academic articles, news coverage

Pre-COVID-19 (before 2020): 12 passages (Egya: 10, Shittu: 2). Post-COVID-19 (2020 and after): 4 passages (Egya: 0, Shittu: 4).

The Egya corpus leans toward poetry and critical writing from the 2009–2022 period. Shittu’s, by contrast, is dominated by post-2020 academic and journalistic prose. Both authors appear in the corpus without prior consent. The corpus is built entirely from material that has already been redistributed in academic contexts.

3.2 Language markers per author

Across the 10 Egya passages, 47 occurrences of culturally loaded vocabulary were catalogued: Igbo terms (chi 13, ala 12, ndi 10, ile 10, ife 6, ani 7, nna 1, ada 1) and Yoruba terms (omo 3, ola 10, iya 1, ifa 1). These are not ornamental; they carry semantic weight (chi = personal destiny/spirit, ala = land-as-ancestral-presence, ndi = communal plural prefix). The 6 Shittu passages contain no Igbo or Yoruba terms; his vocabulary is academic-theoretical (representation, narrative, articulation, interrogation, discourse) plus institutional Nigerian references (ANA, Association of Nigerian Authors). This asymmetry is preserved in the LLM imitation prompts: each model received the same exemplar passage and the same per-sample topic seed, but the system prompt described the target author’s stylistic markers.

4. LLM Imitations

4.1 Generation setup

Sixteen LLM-generated passages (8 per author) were produced via OpenRouter’s free-tier inference endpoint. Two models were used: `nex-agi/nex-n2.5-mini:free` and `nex-agi/nex-n2.5-pro:free`. Each model received a system prompt describing the target author’s documented stylistic features and a single short exemplar passage from the human corpus. Per-passage user prompts were ten topic seeds rotated across calls. Generation temperature was set to 0.01 (near-deterministic sampling) to maximise reproducibility across the two free-tier providers. Max tokens: 600.

4.2 Output inventory

Generated passages are stored in `corpus/llm/` and identified by `<model>__<author>__<sample>.txt`. Total LLM word count: 2,623 (Egya: 814; Shittu: 1,809). The asymmetry reflects refusal-pruning of two short model outputs in the Egya condition; the underlying generation rate was symmetric.

Model	Egya passages	Shittu passages	Total
nex-agi/nex-n2.5-mini:free	5	6	11
nex-agi/nex-n2.5-pro:free	4	1	5

Two additional free-tier models (`poolside/laguna-s-2.1:free` and `nvidia/nemotron-3-nano-omni-30b-a3b-reasoning:free`) were tested but produced rate-limit errors during the generation window. Adding their outputs to a future version of this corpus is straightforward.

5. Features

Nine stylometric features were extracted per passage using `pipeline/analyze_basic.py`:

Feature	Definition
<code>n_words</code>	Word count, after tokenisation
<code>n_sents</code>	Sentence count, split on <code>[. ! ?] \s+</code>
<code>mean_sentence_len_chars</code>	Mean sentence length in characters
<code>sent_len_std</code>	Population standard deviation of sentence length
<code>ttr</code>	Type-token ratio over lowercased word tokens
<code>mean_word_len</code>	Mean word length in characters
<code>punct_density_per_word</code>	Count of <code>. , ; : ! ? () [] { } \ " ' </code> per word
<code>function_word_ratio</code>	Fraction of tokens in a 64-word English function-word list
<code>mean_syllables_per_word</code>	Vowel-group syllable estimate per word

These features are intentionally lightweight. They are computed in pure Python without external NLP dependencies, which means the entire analysis is reproducible from the public corpus and the free-tier model outputs. Pure `stdlib`. Nothing hidden.

6. Results

Per-passage features are in `results_v9/passages.csv`. Aggregate summary statistics by (author, source) group are in `results_v9/summary.json`. The headline findings:

Group	n	sent_len	sent_std	ttr	word_len	fw_ratio	syl/word
egya_human	10	151.0	114.5	0.594	4.62	0.394	1.65
egya_llm	8	210.5	109.0	0.767	4.37	0.402	1.44
shittu_human	6	146.7	109.4	0.542	5.30	0.318	1.94
shittu_llm	8	127.8	66.5	0.642	4.79	0.407	1.73

Two robust signals distinguish human from LLM.

6.1 Passage length, and type-token ratio

The two conditions are not length-matched, and this matters for everything downstream. Human passages average roughly 450 words (range 600-880 for the essays; several shorter lyric passages pull the mean down); LLM imitations average roughly 164 words (range 24-166 under a 600-token generation cap, with refusal-pruning removing two short outputs). The distributions barely overlap.

Group	n	mean words/passage
Egya human	10	~560
Egya LLM	8	~102
Shittu human	6	~502
Shittu LLM	8	~226

TTR falls mechanically as passages get longer, because repeat probability grows with length. A 100-word passage scoring 0.77 is not lexically “wider” than a 560-word passage scoring 0.59; the shorter text has simply not run out of new words yet. The raw TTR gap (Egya +0.17, Shittu +0.10 for the LLM side) is therefore reported as a **confound, not a finding**. No length-matched comparison (fixed-window TTR, MATTR, or matched subsampling) has been run at this corpus size; until one is, the TTR numbers should not be cited as an imitation signature.

6.2 Mean syllables per word

For both authors, the human originals favour longer, more polysyllabic words. Egya: 1.65 against LLM 1.44. Shittu: 1.94 against LLM 1.73. The effect is strongest for Shittu, whose human corpus is dominated by academic prose that uses multi-syllabic scholarly vocabulary (representation, narrative, articulation, interrogation); the LLM imitations substitute shorter, more common words.

Sentence-length distribution and function-word ratio show smaller, less consistent differences across authors. Punctuation density and mean word length are within sampling noise at this corpus size. The two signals above are the load-bearing findings.

7. Discussion

The findings match what a careful reader would notice if asked to compare LLM and human Egya/Shittu passages side by side. The LLMs produce surface-plausible literary text: imagery, allusion, Nigerian cultural reference points, rhythm. The lexical strategies differ. Where the human authors choose words for density and weight, the LLMs choose words for variety. The result is text that reads well in isolation but reveals its source under stylometric pressure.

I treat these results as an observational note on imitation artefacts, not as evidence that authorship attribution is or is not forensically defensible in the LLM era, with three caveats. The headline caution is length: TTR is the most length-sensitive feature in the set, the two conditions differ $\sim 3\times$ in average passage length, and no length-matched comparison has been run, so the TTR result should be read as a confound, not a finding. The syllable-length gap is the load-bearing observation.

7.1 Corpus size

First, the corpus size (16 passages per condition) is small. The syllable-length difference is large enough that it survives multiple-testing correction: with six measured features per group, a Bonferroni-corrected $\alpha = 0.05/6 \approx 0.0083$ still allows rejection for the syllable gap. The TTR difference is reported but not relied on, because TTR is length-dependent and the two conditions differ in length by roughly a factor of three. A larger, length-matched corpus is needed before either effect size can be estimated with confidence intervals.

7.2 LLM provider coverage

Second, only two LLM providers were tested. The two `nex-agi` models produced the bulk of the LLM corpus. Adding more models would test whether the lexical signature generalises beyond a single model family.

7.3 Feature depth

Third, the features used are deliberately lightweight. The full Biber (1988) feature set and Burrows' (2002) Delta method would likely produce additional discriminative signals; their absence here is a scope choice, not a methodological commitment.

8. Future Work

Three directions. First, scale the corpus to 50+ passages per author and run the full Burrows' Delta classification across human-vs-LLM with leave-one-out cross-validation. Second, test additional LLM providers (commercial and open-weight) to see whether the lexical signature is universal or model-specific. Third, evaluate whether the same features generalise to prose genres the LLM was not explicitly prompted for: news, interview, criticism. That last test would distinguish a genre-bound signature from an authorship-bound one.

9. Reproducibility

All code, corpus, generation logs, and outputs are public at <https://github.com/Rawbeew/nigerian-forensic-stylistics>.

Key files: - `corpus/`: human corpus (16 `.txt` files + `metadata.csv`) -
- `corpus/llm/`: LLM imitations (16 `.txt` files) -
- `pipeline/generate_llm_imitations.py`: LLM generation script -
- `pipeline/analyze_basic.py`: stylometric extractor -
- `pipeline/stylometric_pipeline.ipynb`: Colab-friendly notebook -
- `results_v9/passages.csv`: per-passage feature table -
- `results_v9/summary.json`: aggregate summary statistics - `paper_v9.md` and `paper_v9.pdf`: this paper

Running the analysis end-to-end takes under a minute on a laptop. The notebook is provided for users who prefer a Colab-based workflow.

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